

MATH20250 HW1 Observations

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Problem-Specific Observations

The following are some specific problems where I identified areas of confusion. Not included are the problems covered in the homework debrief – for those, consult the debrief file on Canvas.

- **LADW 1.1.2**

- (a) **On the interval $[0, 1]$:** This means $[0, 1]$ is the *domain* of the function, not the range. In this instance, the range can be assumed to be $\mathbb{R}$. Checking the axioms, we see that the set of all continuous functions on $[0, 1]$ **is** a vector space.
- (b) **The set of all polynomials of exactly degree n :** Let $n \in \mathbb{N}$ (necessarily $n > 0$), and consider $f(x) = 2x^n$ and $g(x) = -2x^n$. We see that

$$(f + g)(x) = 2x^n - 2x^n = 0 \Rightarrow \deg((f + g)(x)) = 0 \neq n.$$

So, this **is not** a vector space. (For this same reason, LADW 1.1.3 (d) is **false**.)

- **LADW 1.2.3; LADW 1.2.4**

- (a) **Basis for space of *symmetric* matrices:** Note that any basis vector for a particularly specified space must *itself* be a vector in that space. So, all of the basis vectors (matrices) for a space of symmetric $n \times n$ matrices must themselves be symmetric and square. Taking $n = 3$ as an example, we have the following as a valid basis:

$$\left\{ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \right\}.$$

- (b) **Basis for a space of *antisymmetric* matrices:** Recall that an $n \times n$ matrix A is *antisymmetric* if $A^T = -A$. This means:
 - i. Its diagonal elements – i.e., all a_{ij} where $i = j$, $1 \leq i, j \leq n$ – must be their own negatives. This means that all diagonal elements must equal 0.
 - ii. Its off-diagonal elements – i.e., all a_{ij} where $i \neq j$, $1 \leq i, j \leq n$ – must satisfy $a_{ij} = -a_{ji}$.

Taking $n = 3$, we have the following as a valid basis:

$$\left\{ \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \right\}.$$

General Observations

These are more general, "be careful"-type observations; most of them are proof-related, though there are a few others worth noting.

- Observations on Proofs

- (a) **Defining variables and other quantities:** Whenever you introduce a new quantity, be it a scalar, vector, etc., **state the space in which it lives**.

Ex. $\mathbf{v} \in V$ (where V is a vector space), $\alpha \in \mathbb{C}$.

The $\in$ symbol, meaning "is an element of," is generated by `\in`.

- (b) **Axioms:** When proving a statement axiomatically, it is best practice to use the axiomatic statement *strictly as stated*.

For example, in the vector space axioms, the only operations defined are **vector addition** and **scalar multiplication**. So, for vectors $\mathbf{v}$ and $\mathbf{w}$ in a vector space V ,

we prefer $\mathbf{v} + (-\mathbf{w})$ over $\mathbf{v} - \mathbf{w}$.

Even though those statements are equivalent in all practical vector spaces, the second requires that subtraction be defined as an operation, which it is not.

- (c) **Justifications:** There are two main things to remember when justifying the steps in a proof:
 - i. **Prove all facts on which you rely.** If you need to use a result that you did not already prove in class, **prove it explicitly**. If you need to use a result that comes from another homework problem, **prove it explicitly**. Do not take other homework problems for granted.
 - ii. If you are unsure whether to justify something explicitly, **do it**. Better safe than sorry.

- Miscellaneous

- (a) **Read your matrix dimensions carefully.** Make sure you do not write down a 3×2 matrix when you are working in $M_{2 \times 3}(\mathbb{R})$.
- (b) **Check your transformations.** Make sure you are sending your vectors to the proper space (i.e., $\mathbb{R}^2$, $\mathbb{R}^3$, etc.). You can check this by multiplying test vectors or counting dimensions.